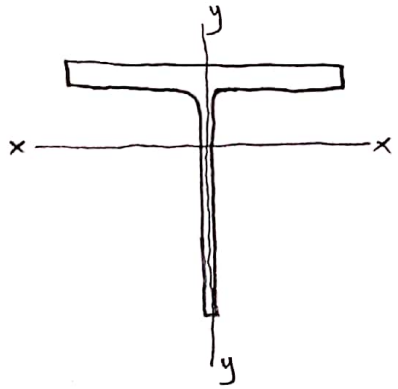


$\frac{1}{2}$IPN400 profilinden oluşturulmuş basınç çubuğunun tasarım basınç dayanımını hesaplayınız.

Compute the compression strength of the compression member which formed by 1/2 IPN400 profile.

$L_{kx} = 400$ cm , $L_{ky} = 200$ cm , $L_{kz} = 200$ cm

1/2 IPN400

$I_x = 2140$ cm^4
$I_y = 578$ cm^4
$i_x = 6,02$ cm
$i_y = 3,13$ cm
$b = 155$
$t_w = 14,4$
$t_f = 21,6$
$d = 200$
$A = 58,9$ cm^2
$e_x = 5,46$ cm
$J = 51,3$ cm^4
$C_w = 490000$ cm^4

S275

$F_y = 27,5$ kN/cm^2
$F_u = 43$ kN/cm^2
$E = 20000$ kN/cm^2
$G = 7720$ kN/cm^2

$\frac{b}{2t_f} = \frac{155}{2 \cdot 21,6} = 3,59 < \lambda_r = 0,56\sqrt{\frac{E}{F_y}} = 15,10$ → non-slender (narin değil)

$\frac{d}{t_w} = \frac{200}{14,4} = 13,9 < \lambda_r = 0,75\sqrt{\frac{E}{F_y}} = 20,23$ → non-slender (narin değil)

→ Flexural buckling for the axis of no symmetry
(Simetrik olmayan eksene göre eğilmeli burk.lma)

$\lambda_x = \frac{L_{kx}}{i_x} = \frac{400}{6,02} = 66,44 < \lambda_r = 4,71\sqrt{\frac{E}{F_y}} = 127,02$ → inelastic buckling (elastik olmayan burk.lma)

$F_e = \frac{\pi^2 E}{\lambda^2} = \frac{\pi^2 \cdot 20000}{66,44^2} = 44,72$ kN/cm^2

$F_{cr} = 0,658^{27,5/44,72} \cdot 27,5 = 21,26$ kN/cm^2

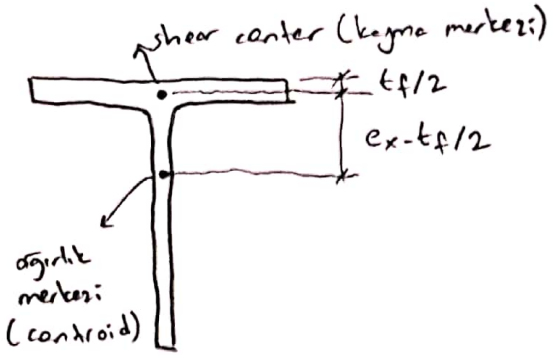
$P_n = 58,9 \cdot 21,26 = 1252,21$ kN

CamScanner ile tarandı

→ Flexural-torsional buckling for the axis of symmetry
(Simetri eksene göre eğilmeli-burulmalı burkulma)

$$F_e = \frac{(F_{ey}+F_{ez})}{2H}\left[1-\sqrt{1-\frac{4F_{ey}F_{ez}H}{(F_{ey}+F_{ez})^2}}\right]$$

$$H = 1-\frac{x_o^2+y_o^2}{\bar{r}_o^2} \qquad \bar{r}_o^2 = x_o^2+y_o^2+\frac{I_x+I_y}{A}$$



$x_o = 0$

$y_o = e_x - \frac{t_f}{2} = 5,46 - \frac{2,16}{2} = 4,38$ cm

$\bar{r}_o^2 = 0+4,38^2+\frac{2140+578}{58,9} = 65,33$

$H = 1-\frac{4,38^2+0^2}{65,33} = 0,7063$

$F_{ey} = \frac{\pi^2 E}{\lambda_y^2} = \frac{\pi^2 . 20000}{63,89^2} = 48,36$ kN/cm² $< \lambda_r = 127,02$ → inelastic buckling (elastic olmayan burkulma)

$$F_{ez} = \left(\frac{\pi^2 . E . C_w}{K_{kz}^2}+G.J\right)\frac{1}{A.\bar{r}_o^2} = \frac{7720.51,3}{58,9.65,33} = 102,92 \text{ kN/cm}^2$$

$$F_e = \frac{(48,36+102,92)}{2.0,7069}\cdot\left[1-\sqrt{1-\frac{4.48,36.102,92.0,7063}{(48,36+102,92)^2}}\right] = 40,56 \text{ kN/cm}^2$$

$F_{cr} = 0,658^{27,5/40,56} . 27,5 = 20,70$ kN/cm²

$P_n = 58,9 . 27,5 = 1219,56$ kN

$\phi P_n = 0,90 . 1219,56 = \underline{1097,61 \text{ kN}}$

CamScanner ile tarandı